MATLAB R2016a

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- #jungkook #jk.jpeg
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- 1fef2246-0411-4ebf-a8...
- 8f9573c9-65b1-49b3-b...
- 24c81b04-5d55-4d62-8...
- 250b5969-d52a-4607-8...
- 461f5fc0-0c6e-4309-9c...
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- 230301 _ taehyung - D' ...
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- A01.m
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Details

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Editor - C:\Users\moni2\Downloads\SASexp15.m

SASexp13.m | SASexp15.m

```
1 -  clc; clear; close all;
2 -  N = 10;                    % rectangular pulse length
3 -  n = 0:N-1;
4 -  x = ones(1,N);
5 -  w = linspace(-pi, pi, 1024);   % dense frequency grid (one period)
6 -  X = zeros(size(w));
7 -  for k = 1:length(w)
8 -      X(k) = sum(x .* exp(-1j*w(k)*n));   % direct DTFT summation
9 -  end
10 -  figure;
11 -  subplot(2,1,1); plot(w/pi, abs(X)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
12 -  title('DTFT Magnitude Spectrum'); grid on;
13 -  subplot(2,1,2); plot(w/pi, angle(X)); xlabel('\omega / \pi'); ylabel('\angle X(e^{j\omega})');
14 -  title('DTFT Phase Spectrum'); grid on;
15
16    %% Periodicity check over 2 periods
17 -  w2 = linspace(-2*pi, 2*pi, 2048);
18 -  X2 = zeros(size(w2));
19 -  for k = 1:length(w2), X2(k) = sum(x .* exp(-1j*w2(k)*n)); end
20 -  figure; plot(w2/pi, abs(X2)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
21 -  title('DTFT Magnitude over Two Periods (Periodicity = 2\pi)'); grid on;
22
23    %% Time-shift property verification
24 -  n0 = 3;
25 -  x_shift = [zeros(1,n0), x];
26 -  n_shift = 0:length(x_shift)-1;
27 -  Xs = zeros(size(w));
28 -  for k = 1:length(w), Xs(k) = sum(x_shift .* exp(-1j*w(k)*n_shift)); end
29 -  mag_err = max(abs(abs(Xs) - abs(X)));
```

Workspace | Command Window

```
Time-shift property: max magnitude error = 7.98e-15
Time-shift property: max phase-term error (rad) = 3.142
Linearity property: max complex error = 2.809e-15
fx >>
```

Ln 48 Col 1

23°C Light rain

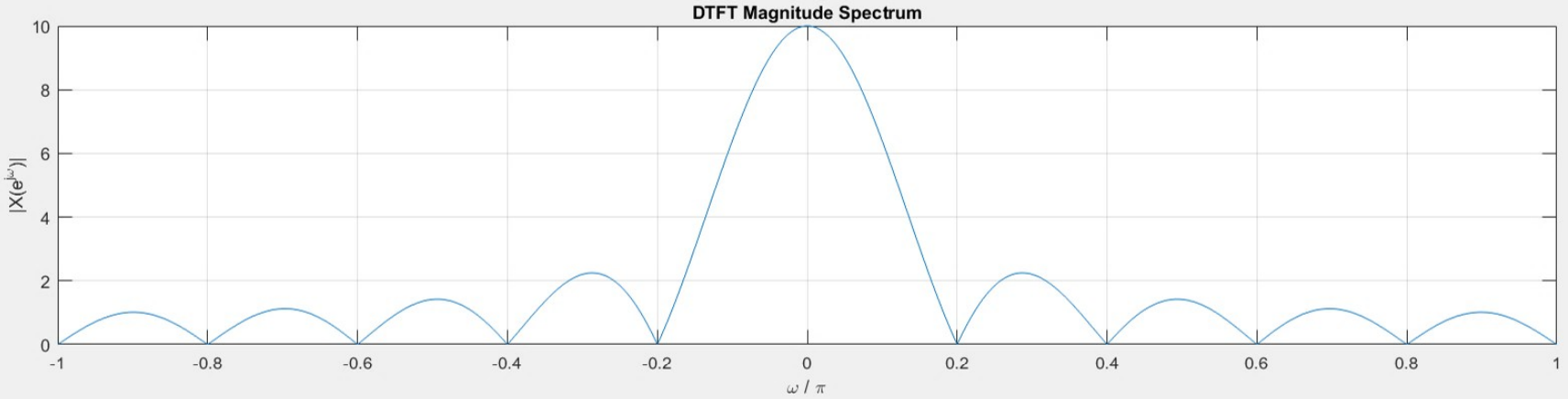
Search

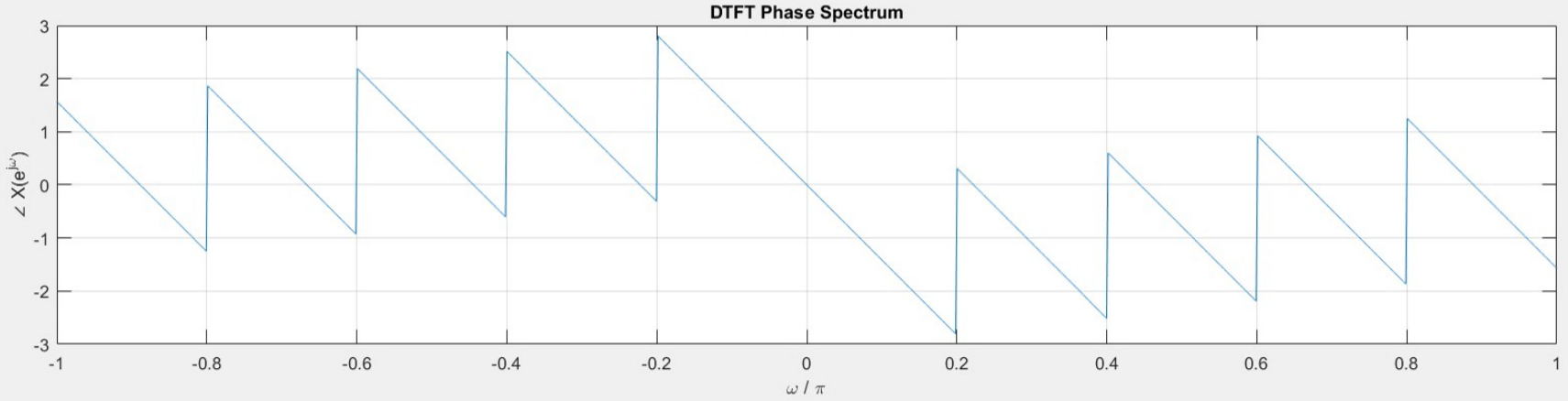
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Figure 1

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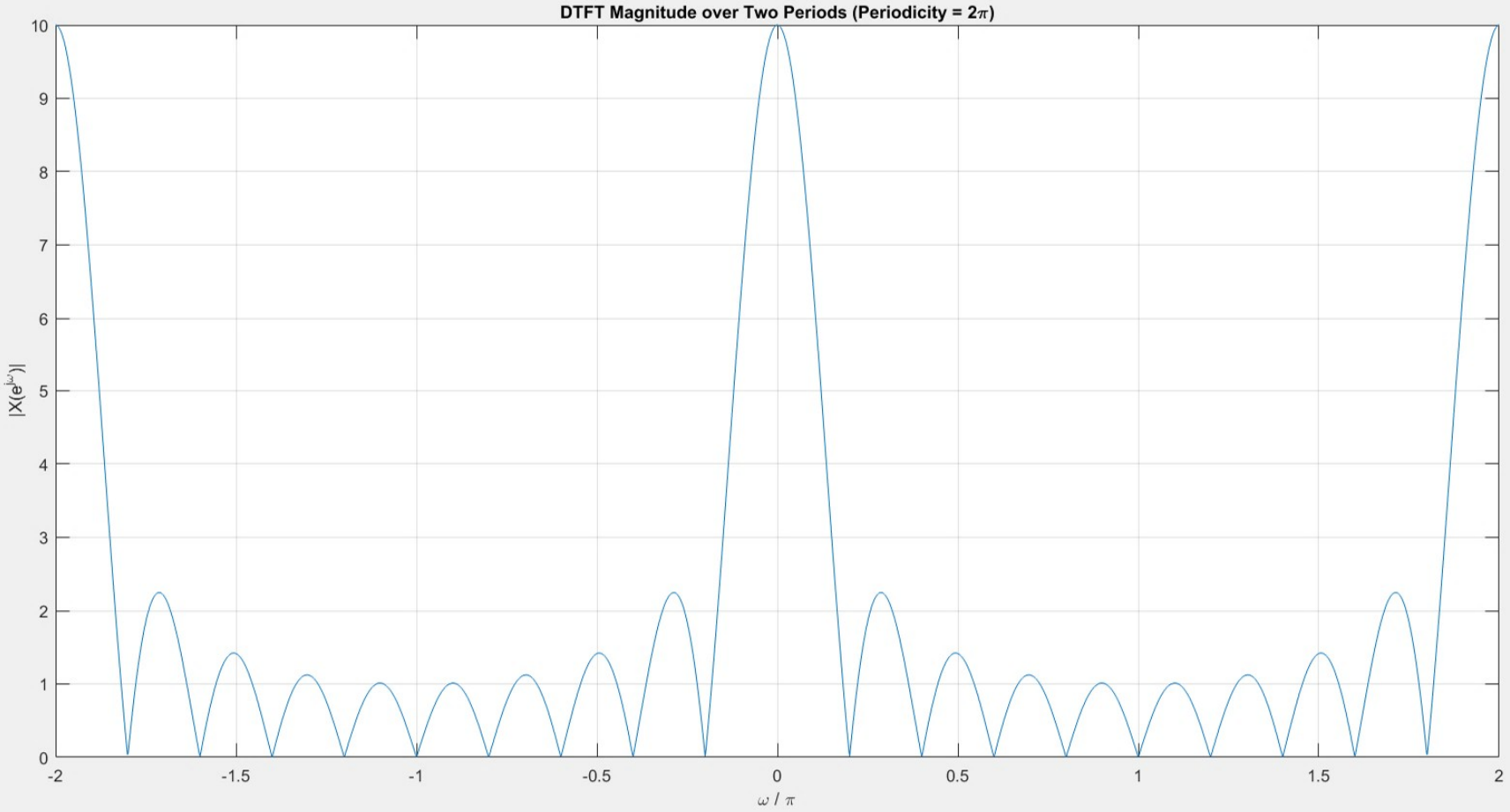




23°C
Light rain
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Figure 2

File Edit View Insert Tools Desktop Window Help



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